CDS6214
Data Science Fundamentals
Analyzing Weather Parameters and Their Impact on Human Comfort in Toronto
Project (40%)
Tutorial Section: TT7L
Group Number: G17

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Task Distribution

Lecture Section: TC3L

Tutorial Section: TT7L

Group Number: G17

Domain: Climate and Environment

Task Distribution:

Student Name	Task Done (Be very specific)
TEOH KAI LOON	Data preprocessing,data transformation,data exploration,data mining,report(motivation,problem,impact on the proposed problem)
WONG KAH CHUN	Data preprocessing,data transformation,data exploration,data mining, report(impact on proposed problem,challenges encountered and limitation)
LAM ZI FONG	Data preprocessing,data transformation,data exploration,data mining, report(Description and implementation)
CHAI DI SHENG	Data preprocessing,data transformation,data exploration,data mining, report(Description and implementation)

Motivation, Problem

The main problem involves studying several aspects of weather parameters and their impact on human comfort in Toronto during 2020.There are several motivations for this analysis.

The first motivation is to ensure public health and safety.Climate plays a crucial role in determining health hazards within the general population.Cold weather poses some dangers,including a high probability of falls due to slippery floors,icy surfaces or even snow. By analyzing these patterns,the risk factors can be understood and ways of reducing them can be developed by the public health officials.

The second motivation is to prevent a negative impact on the economy. Weather parameters can influence the local economy,especially the retail,hospitality and transportation industries. Knowledge of these patterns can therefore aid a business in planning for change in demand and managing working capital effectively.

The third motivation is to help in urban planning and infrastructure.Climate analysis helps to predict the impact on comfort of individuals and it can be used in city planning and construction innovation.This data can used by city planners to apply better architectures in building construction and the design of public areas and infrastructure such as transportation so that it will not deteriorate rapidly when faced with adverse weathers.

The last motivation is to improve comfort and quality of life.With this way,we can get insights about how the weather parameters influence human comfort and be able to progress in enhancing well being. Such concerns include advice on what to wear,where to live and what to do in case of certain weather.

Impact on the Proposed Problem to be solved to communities / society / nation

The proposed problem in the report is to study the relationship between every attribute related to the temperature and weather in a region such as the temperature, dew point, wind direction, wind speed, wind chill, humidity, visibilities and weather. Understanding and analyzing the data attributes could bring some significant impact to the community and people

First of all, by utilizing the weather visualization ,the proposed problem can help the agriculture industry to deduce what kinds of vegetables and fruits are suitable for planting in that season. It is also possible to use weather forecasting to build 'shelters' for crops in advance to prevent them from being destroyed by unpredictable weather.

Furthermore, the safety of the population is also guaranteed especially in transportation. By analyzing how humidity affects the visibility to know if it is suitable for driving on the data. For example, a driver's visibility may be impeded by heavy snowfall in winter; the temperature and humidity cause the car's mirrors to fog up during winter months, thus impairing the driver's vision.

In addition, analyzing the temperature and dew point correlations can predict the weather conditions to avoid affecting the daily life of one. People could know the temperature and to prepare an umbrella in advance.

Moreover, studying the variation of wind speed is also suitable to help in the area of energy. Visualize the data to understand whether this location is suitable for building a wind turbine in order to capture wind energy and to generate more energy resources.

Description and implementation of Pipeline solutions for each DS process component.

1.Exploratory Question

Question 1: How do temperature and relative humidity vary on a monthly basis throughout 2020?
Objective: Identify monthly trends and patterns in temperature and humidity to understand fluctuations within the year.

Question 2: How does wind speed vary on a monthly basis throughout 2020?
Objective: Identify monthly patterns in wind speed and dominant wind directions.

Question 3: Is there a discernible relationship between humidity levels and visibility in Toronto City Centre during 2020?
Objective: Understand the potential relationship between humidity levels and visibility in Toronto City Centre during 2020.

Question 4: What is the relationship between temperature and dew point in 2020?
Objective: Understand how temperature and dew point correlate to assess humidity levels.

2. Data Collection

The dataset used for this project is Toronto City Centre Hourly Weather Data 2020 by Jason Ho from Kaggle community. There are 8,784 weather data points recorded from a weather station located on Toronto Island, in Toronto, Ontario, Canada. The CSV contains 28 features.

The weather data includes station location information (longitude, latitude and station name), date and time information (year, month, day, time), weather elements (temperature, dew point temperature, relative humidity, wind direction, wind speed, visibility, station atmospheric pressure), and derived weather indexes (humidex and wind chill). For each weather element, there is also a flag to indicate unique events.

3. Data Preprocessing

During data preprocessing, the weather data is first loaded. A data frame is created to organize the information, including details about each column (names, data types, and the number of missing values). In the data cleaning stage, columns with a high percentage of missing values are typically dropped. For numeric columns like Temperature, Dew Point Temperature, Relative Humidity, Wind Speed, Visibility, Station Atmospheric Pressure,

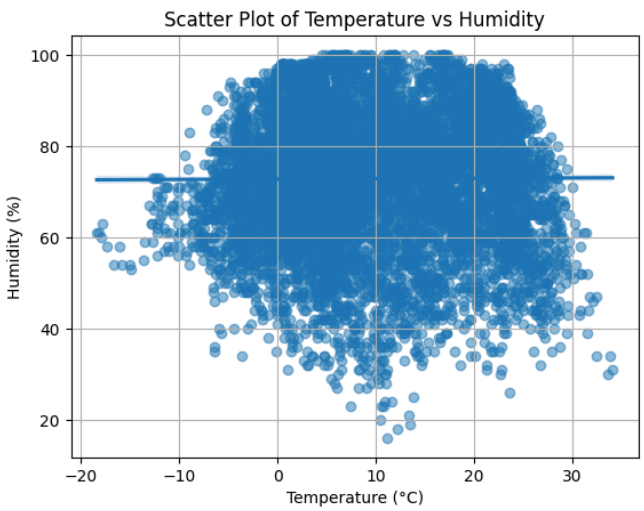
missing values can be imputed using techniques like mean imputation. Then, features that are not relevant to the analysis are removed.

Data transformation is then applied to categorical data, such as time strings. A technique called label encoding is used for this purpose. Then, the data is divided into two sets: a training set used to build the model and a testing set used to evaluate the model's performance.

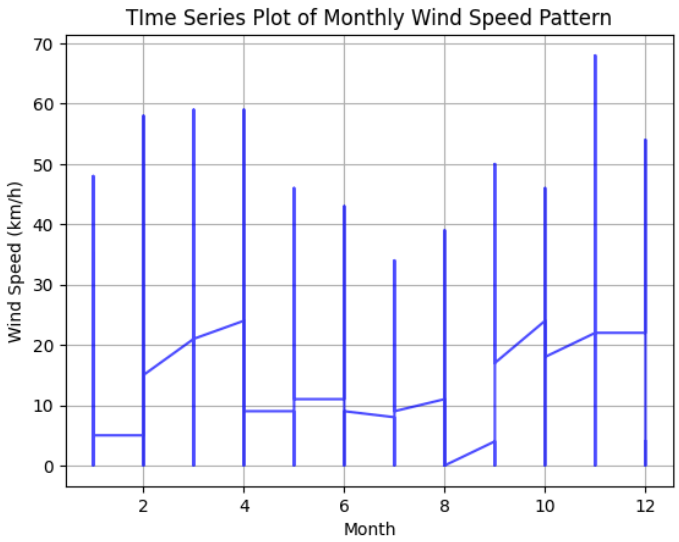
4. Data Exploration and Visualization

Data exploration and visualization involve examining data attributes for the relationships and trends, deriving value-added insight through picture based tools. Utilizing pandas for data transformation, this process begins with loading and reading cleaned and processed raw data. Exploratory Data Analysis (EDA) is then executed, including the calculation of summary statistics, testing for correlations and the preparation for graphs. Visualization is performed by plotting histograms, scatter diagrams, pie-charts or any other necessary diagram to illustrate the relationships or the patterns that may exist in the data.

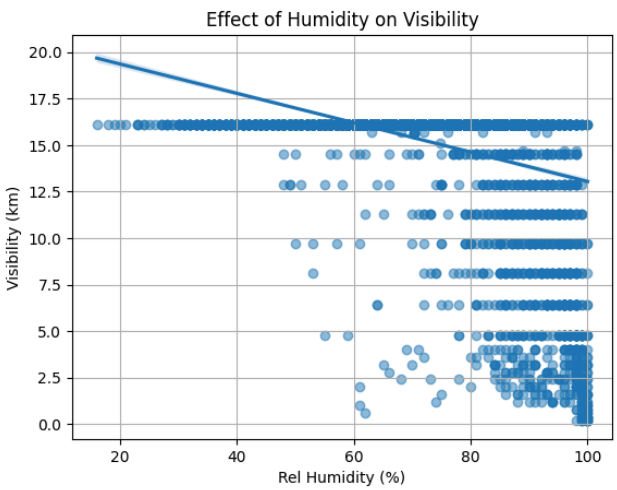
Based on the scatter plot below, it shows horizontal bands of points, indicating that for certain temperature ranges, there is a wide variability in humidity levels. There is a high density of points around 0 to 10°C and 60% to 80% humidity, and some points that appear to be outliers, particularly at lower humidity levels (below 20%) and extreme temperatures (below -10°C and above 30°C). We can conclude that the data points are spread across a wide range of humidity levels for most temperature values, indicating no clear linear relationship between temperature and humidity.



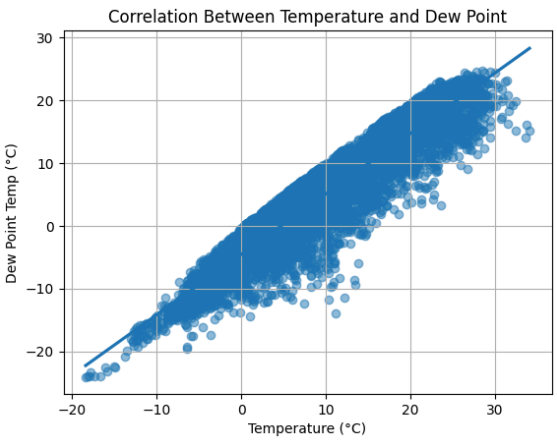
From the time series plot of the Monthly Wind Speed pattern, we can observe that the wind speed shows significant month-to-month variability. On some months, the wind speed is relatively low (below 10 km/h), while on other days, it peaks much higher (up to around 50 km/h). We can conclude that the wind speed appears to have no clear periodic patterns or trends over the 12-month period.



For the graph below, there is a clear negative correlation between relative humidity and visibility. As humidity increases beyond 60%, visibility starts to decline more noticeably. When humidity reaches about 80% to 100%, visibility often drops to below 15 km, with many instances even falling below 10 km.



The graph below is displaying the relationship and correlation between temperature and dew point in the city of Toronto city. The correlation between them has a very high positive correlation. This indicates that the two variables are strongly linearly related with each other. When the temperature is increased, the dew point temperature tends to increase also.



5.Data Mining

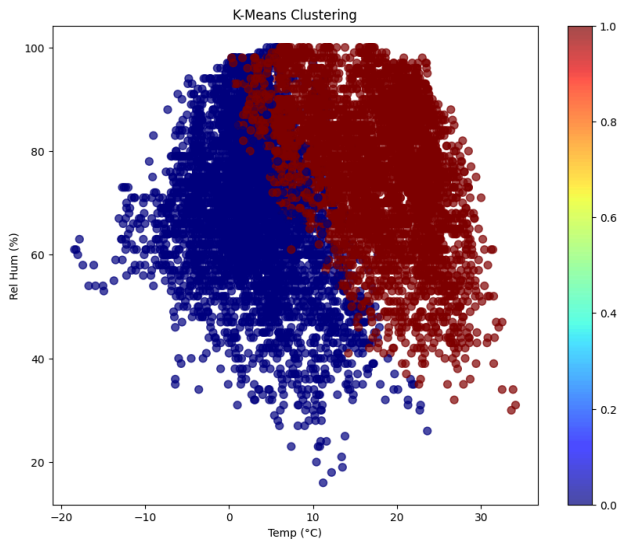
Lastly, data mining is applied by incorporating several developed statistical and machine learning methods, which is used to find out all related information, consistent as well as associations of the data. Some techniques can be used by data mining such as data grouping, association, identifying outliers, data time-series feature and so on. We can first apply the appropriate data mining algorithms depending on the type of data collected and goals of the enterprise. Then, analyzing the data mining algorithms outputs to arrive at the knowledge level to interpret the result. And lastly, iterate and reuse the methods based on preliminary results, with a view of identifying additional features and patterns.

First, we try the K-Mean Clustering for our exploratory question to observe the data patterns and set the number of clusters to two.

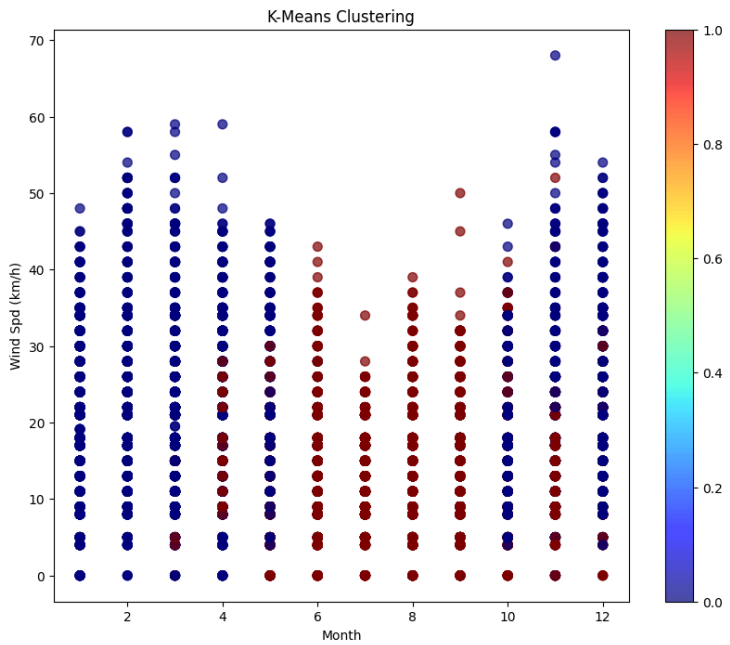
```
[ ] model = KMeans(2, verbose=0, n_init='auto')
    model.fit(X)
```

```
KMeans
KMeans(n_clusters=2, n_init='auto')
```

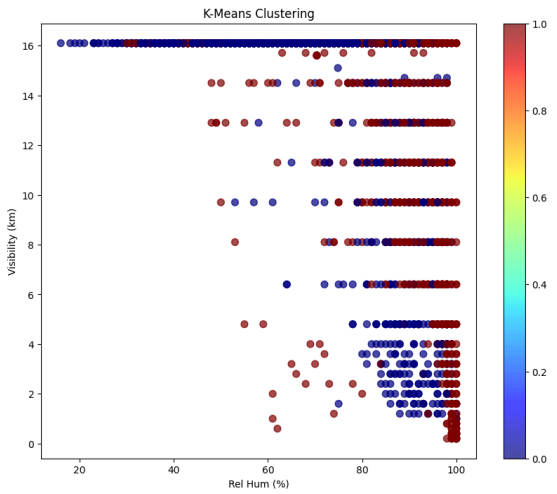
Based on the graph below, we can observe that the clusters appear to be somewhat well-separated along the temperature axis. The blue cluster is generally associated with lower temperatures, while the red cluster is associated with higher temperatures. Both clusters have a significant overlap in the relative humidity range, particularly between 40% and 100%. However, the blue cluster tends to have more data points in the lower humidity range, whereas the red cluster does not have many points in this lower range.



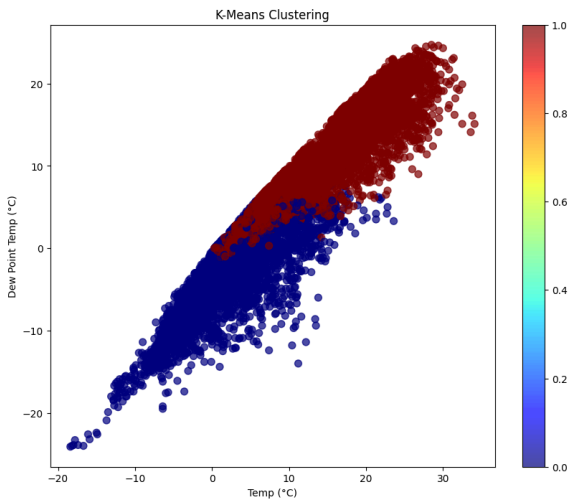
For the graph below, the clusters seem to be distributed across the month without a clear temporal pattern. Both clusters are present most months, indicating no significant temporal separation between the clusters. We can observe the wind speed patterns based on the blue cluster having more data points at higher wind speeds (above 60 km/h) while the red cluster has more data points at lower wind speeds (below 54 km/h).



The graph below shows that the blue cluster primarily consists of data points with high visibility (around 16 km) while the red cluster has data points distributed across a wider range of visibility values. The red cluster includes data points with lower visibility (less than 2km) and higher relative humidity (above 80%). This suggests that lower visibility is associated with higher humidity in the red cluster.



From the graph below, there appears to be a strong positive correlation between Temperature and Dew Point Temperature, as indicated by the general upward trend of the data points in both clusters. It means that as the temperature increases, the dew point temperature also tends to increase.



After this, we also try to employ a Random Forest model to predict the future temperature. The evaluation metrics we are using are Mean Absolute Error (MAE), Mean Squared Error (MSE), and R^2 Score and the result is shown below. For the result of Mean Absolute Error (MAE), it shows that the model's predictions are off by approximately 0.1256 units from the actual values and this is a relatively low error. Next, for an MSE of 0.0507 indicates a low

variance in the error, which implies that the model's predictions are consistently close to the actual values. Lastly, an R^2 score of 0.9994 suggests that 99.94% of the variance in the actual values is explained by the model. This is a very high R^2 score, indicating excellent model performance and a strong fit to the data. In conclusion, the Random Forest Regressor model demonstrates excellent predictive performance on the dataset, with low error rates and a high R^2 score. The model accurately captures the relationship between the features and the target variable, making it a reliable tool for future predictions.

```
Model Evaluation Results:
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Mean Absolute Error (MAE): 0.12558714608836613
Mean Squared Error (MSE): 0.05074076526438292
R^2 Score: 0.9994180858723073

Actual vs Predicted Values (First 10 samples):

   Actual Values  Predicted Values
0           15.6            15.434
1            1.9             1.745
2           16.5            16.658
3            3.9             3.819
4           -2.6            -2.467
5           17.0            16.686
6            1.0             1.013
7           21.8            21.792
8           24.1            24.225
9            1.0             1.125
```

Challenges encountered and limitation of current work

Challenges encountered

Navigating the vast amount of data sources available on the internet have become one of the significant challenges for us. We need to consider and ensure the data information such as the validity of the data. We should carefully consider various aspects such as the reliability of data sources. In the end, we have decided to use the data from the Kaggle site, which is now a cloud of Google part, and offer authorized data resources to able users to exchange the code and data set.

Besides that, the data that we finally obtain have a limitation of data integrity. There are some lack of value and missing value for some of the par. This becomes one of the limitations as missing value may lead to an impact of inaccuracy of result output. To address this, we finally replace the missing value by a common method which fills in the mean value for the missing data value of the column.

Limitation of current work

When performing weather analysis, the influence of external factors such as human activities in metropolitan context and climate changes caused by environmental changes and introduce important restrictions. Accurately identifying and analyzing meteorological trends is challenging due to ongoing climatic and natural changes, which can complicate meteorological analysis and reduce the reliability of output results.

References

-Jason ho. (n.d.). Toronto City Centre Hourly Weather Data (2017-2020) [Data set]. Kaggle. Retrieved June 25, 2024, from https://www.kaggle.com/datasets/jasonzxho/toronto-city-centre-hourly-weather-data-20172020

-Ku Day Chyi. (2024). Lab Exercises conducted for [CDS6214-DATA SCIENCE FUNDAMENTALS] at Multimedia University.